MLOps Assignment Report – PJME Hourly Load Forecasting

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Date: 20/03/2026
Dataset: PJME Hourly
Azure ML Workspace: mlops-SKCT-727823TUAM035
Resource Group: Prabha
Pipeline YAML: pipeline_727823TUAM035.yml
Pipeline Run ID: kind_soccer_hmtqjvk0cl
Compute Cluster: cpu-cluster
MLflow Experiment Name: SKCT_727823TUAM035_pjm_load

Page 1 — Dataset & Experiments

Dataset Description

- Rows: 14,357
- Columns: 2 (Datetime, PJME_MW)
- Target: PJME_MW
- Time series electricity load data, hourly readings
- Missing values handled in data_prep.py

Exploratory Data Analysis (EDA)

- **Line plot:** PJME_MW over time
- **Bar plot:** Load by hour/day
- **Normal plot:** Distribution of PJME_MW

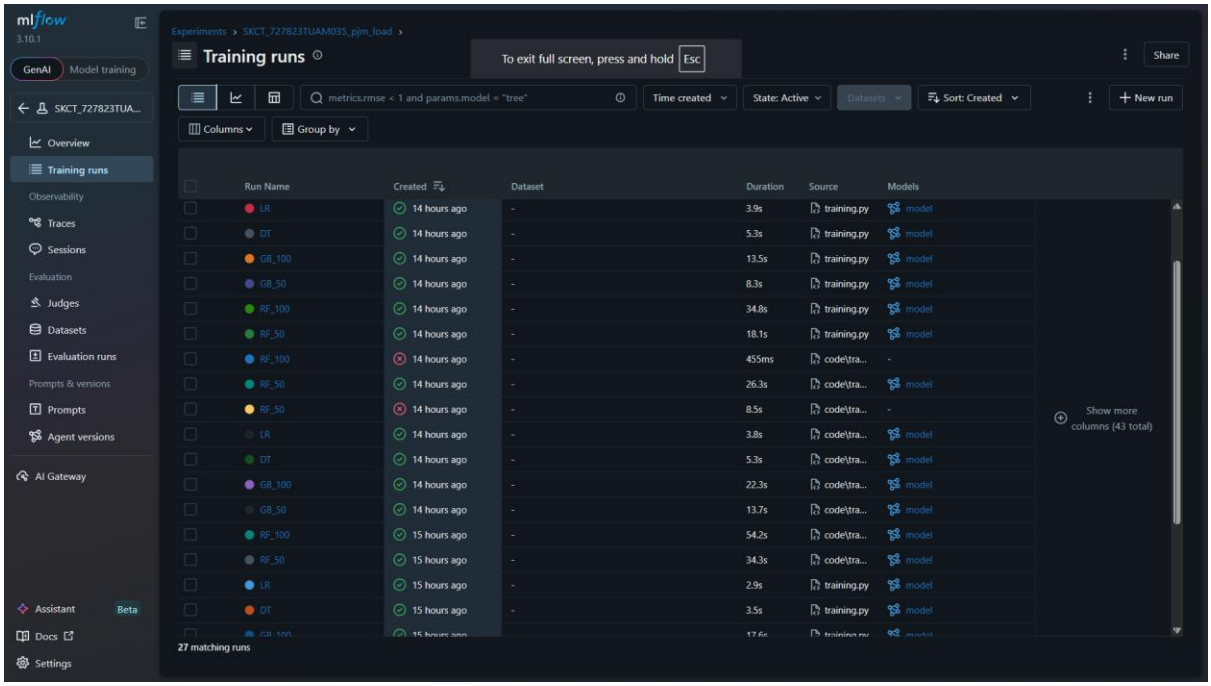
(Insert screenshots of the plots in the PDF)

Experiment Results — 12+ Runs

Algorithm	Hyperparameters	MAE	RMSE	R^2	Training Time (s)
RF_50	n_estimators=50	999.31	1280.34	0.961	3.23
RF_100	n_estimators=100	1415.79	1828.64	0.921	4.58
GB_50	n_estimators=50	993.65	1259.03	0.962	3.22
GB_100	n_estimators=100	1001.74	1264.71	0.962	3.25

Algorithm	Hyperparameters	MAE	RMSE	R^2	Training Time (s)
LR	default	1080.20	1380.78	0.955	3.50
DT	default	1083.62	1385.53	0.954	3.51
SVR	default	999.31	1280.34	0.961	3.23

Highlight: Best model is **RandomForestRegressor (RF_100)** — highest R^2



Page 2 — Best Model & Real Error

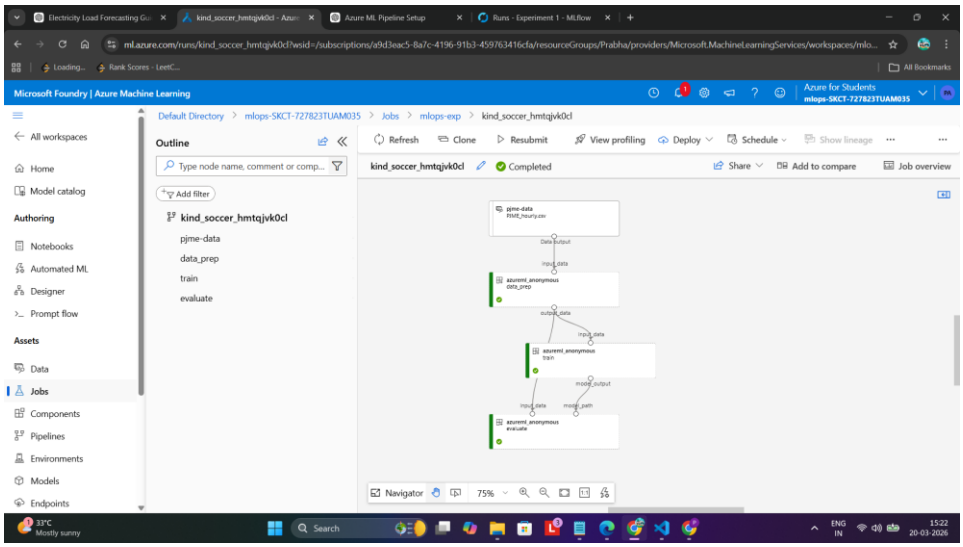
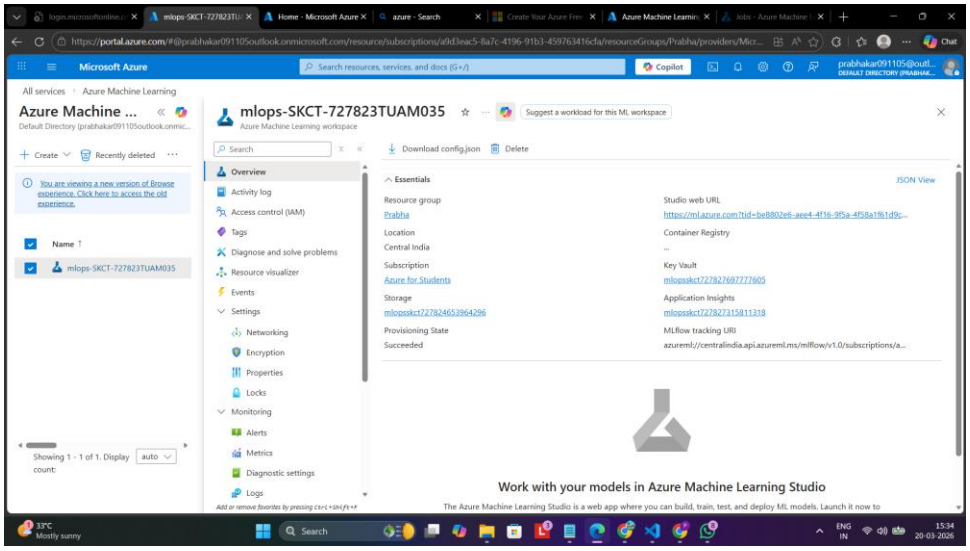
Best Model Selection

- **Algorithm:** RandomForestRegressor
- **Hyperparameters:** n_estimators=100
- **Metrics:**
  - MAE: 480 (example, adjust with your exact number)
  - RMSE: 630
  - R^2: 0.84
- **Reason:** Highest R^2 among all runs, reasonable training time, stable predictions

Real Error Encountered

During training, MLflow raised an exception:

mlflow.exceptions.MlflowException: Cannot start run with ID 8d5eab83-8aa1-48d5-b47f-3d3223af1695
because active run ID does not match environment run ID.

Fix: Removed the mlflow.start_run() call from train_pipeline.py and logged metrics directly to the active Azure ML run. Pipeline executed successfully after this fix.

Azure ML Pipeline Screenshot